

Dat Lam

(647) 906-3849 | lntd6534@gmail.com | linkedin.com/in/dat-lam-26a6b62b2 | github.com/DavidLam0101

EDUCATION

Carleton University

Ottawa, ON

Bachelor of Science in Computer Science (Honours)

- **Relevant Coursework:** Operating Systems, Design and Analysis of Algorithms, Database Management Systems, Discrete Structures, Fundamentals of Web Applications.

TECHNICAL SKILLS

Languages: Java, C++, C#, HTML, CSS, JavaScript, SQL

Frameworks & Libraries: React, Node.js, Spring Boot, PostgreSQL

Developer Tools: Git, GitHub, Linux, GDB

PROJECTS

Nyctophobia | *Godot, GDScript*

- Architected diverse enemy archetypes with time-based dynamic difficulty scaling, alongside a procedural maze generation algorithm that adapts structural complexity to real-time player progression metrics.
- Designed a tiered loot economy with three chest rarities, each governed by weighted probability distributions that scale buff drop rates with cost, encouraging strategic resource allocation.
- Implemented A* pathfinding for responsive enemy navigation through dynamically generated mazes, and developed an autonomous companion AI featuring priority-based target selection and state-driven behavior.
- Produced original character assets to support the game's visual identity.

Cave Diver | *Godot, GDScript*

- Designed and balanced wave-based enemy encounter systems with algorithmic spawn patterns, maintaining consistent difficulty curves across dungeon levels.
- Built an intuitive HUD and menu system with clear visual hierarchy and responsive player feedback, and implemented modular character ability systems.
- Coordinated cross-disciplinary tasks across design, art, and programming pipelines to deliver project milestones on schedule.

Personal Portfolio | *React, JavaScript, Spring Boot*

- Developed a responsive single-page portfolio with smooth scroll navigation, dynamic project filtering, and a mobile-first layout optimized across viewports.
- Built a modular component architecture using React functional components and hooks (`useState`, `useEffect`) for efficient state management.
- Implemented a secure content management feature with a Spring Boot backend using JWT authentication, enabling protected editing of portfolio content.